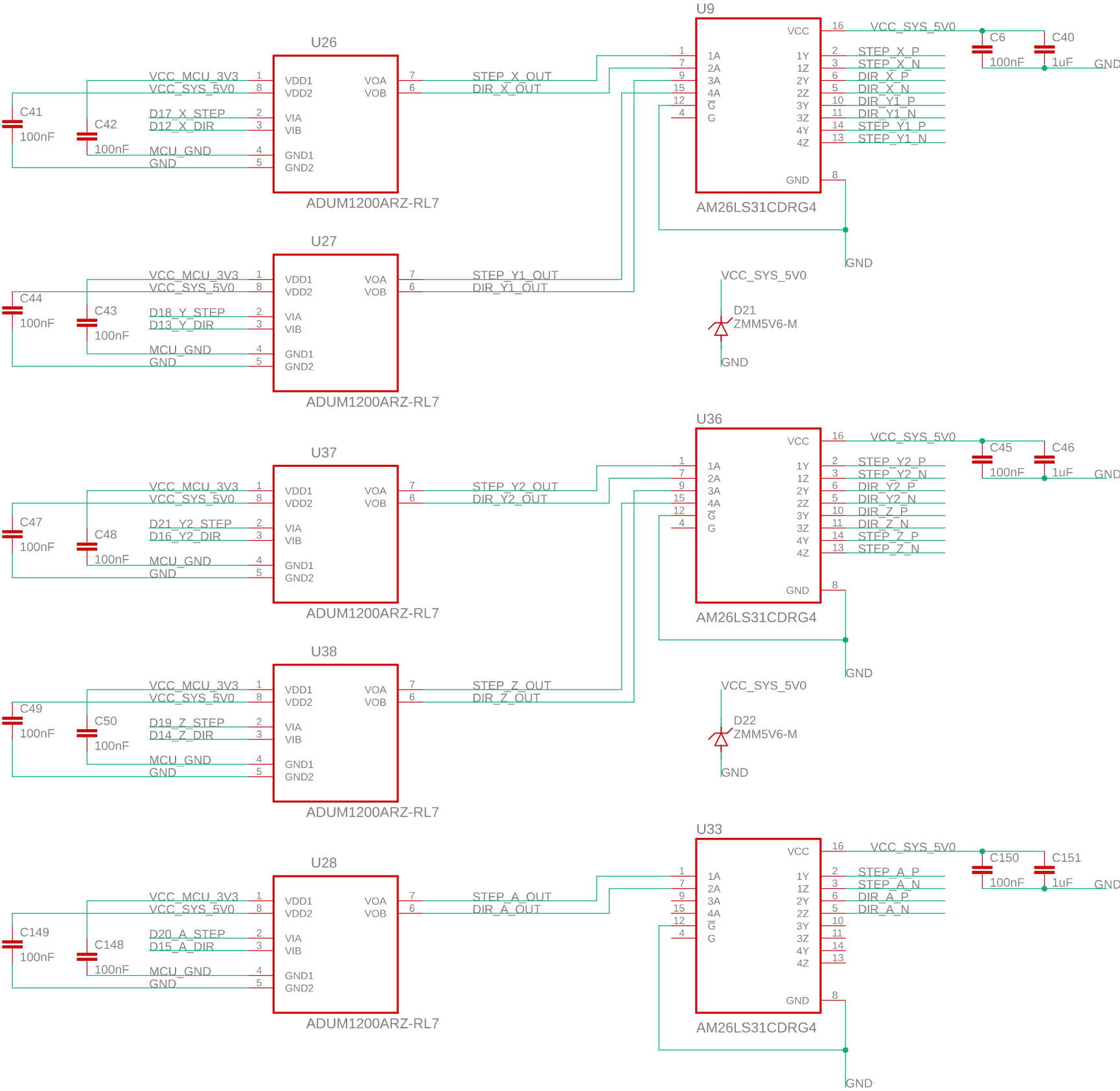


MOTOR STEP DIR BUFFER



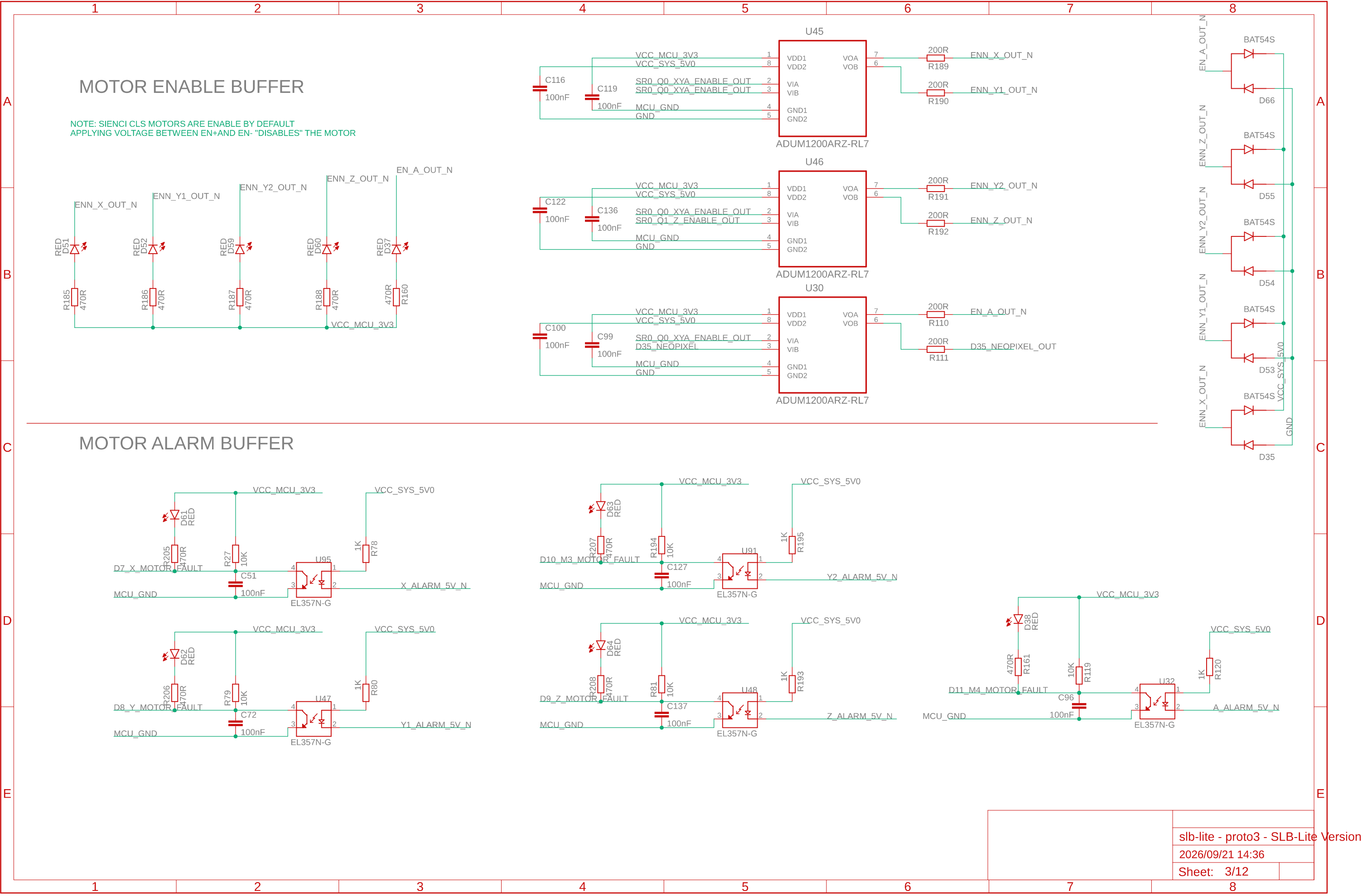
STEP_X_N	7	7	14	X LIM_EXT
STEP_X_P	6	6	13	LIMIT_VCC_SEL
DIR_X_N	5	5	12	GND
DIR_X_P	4	4	11	X_ALARM_5V_N
ENN_X_OUT_N	3	3	10	GND
VCC_SYS_5V0	2	2	9	MOTOR_VCC_24-48V
XYA_HOME_EN	1	1	8	GND

STEP_Y1_N	7	7	14	Y1 LIM_EXT
STEP_Y1_P	6	6	13	LIMIT_VCC_SEL
DIR_Y1_N	5	5	12	GND
DIR_Y1_P	4	4	11	Y1_ALARM_5V_N
ENN_Y1_OUT_N	3	3	10	GND
VCC_SYS_5V0	2	2	9	MOTOR_VCC_24-48V
XYA_HOME_EN	1	1	8	GND

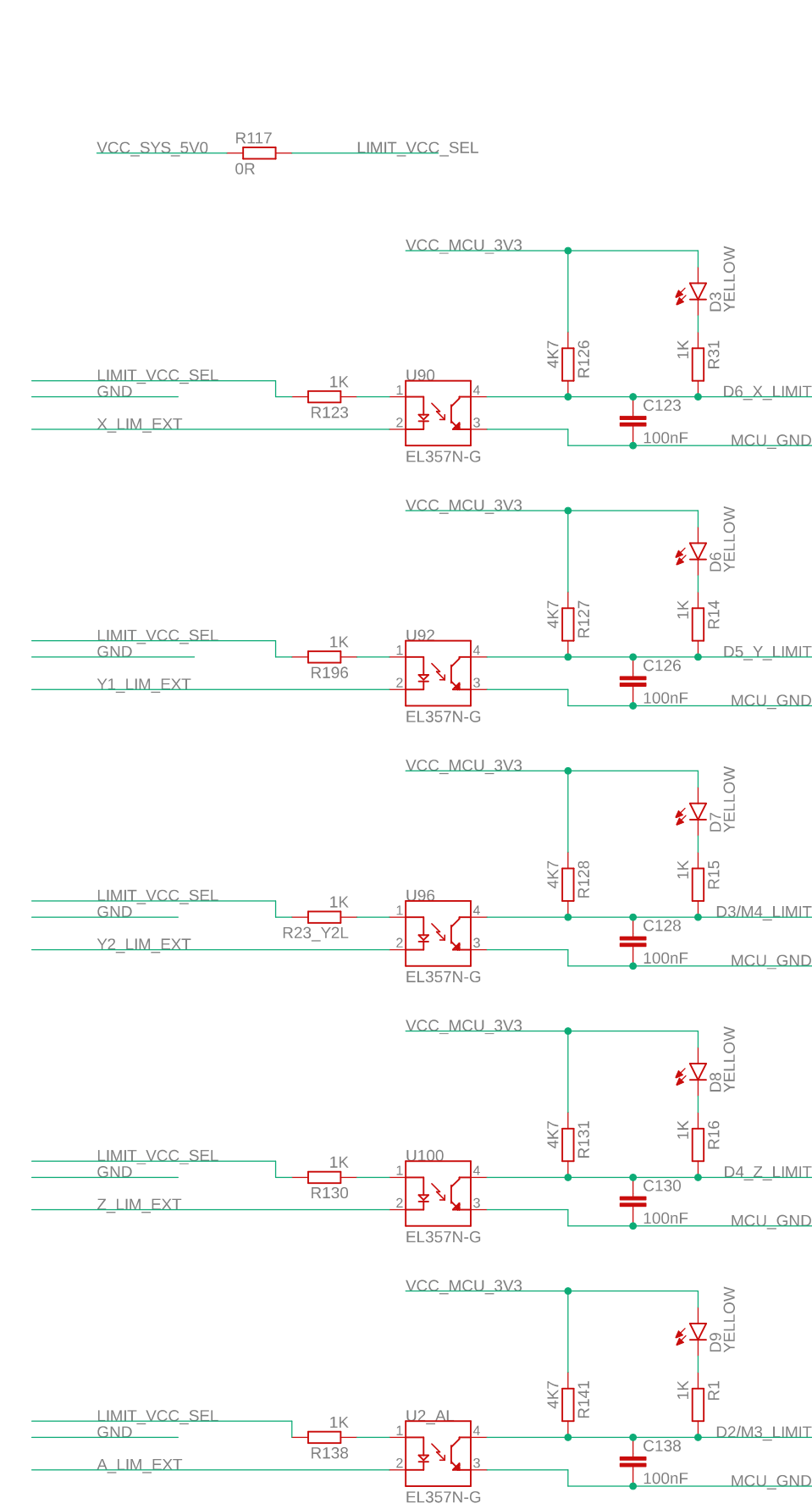
STEP_Y2_N	7	7	14	Y2 LIM_EXT
STEP_Y2_P	6	6	13	LIMIT_VCC_SEL
DIR_Y2_N	5	5	12	GND
DIR_Y2_P	4	4	11	Y2_ALARM_5V_N
ENN_Y2_OUT_N	3	3	10	GND
VCC_SYS_5V0	2	2	9	MOTOR_VCC_24-48V
XYA_HOME_EN	1	1	8	GND

STEP_Z_N	7	7	14	Z LIM_EXT
STEP_Z_P	6	6	13	LIMIT_VCC_SEL
DIR_Z_N	5	5	12	GND
DIR_Z_P	4	4	11	Z_ALARM_5V_N
ENN_Z_OUT_N	3	3	10	GND
VCC_SYS_5V0	2	2	9	MOTOR_VCC_24-48V
Z_HOME_EN	1	1	8	GND

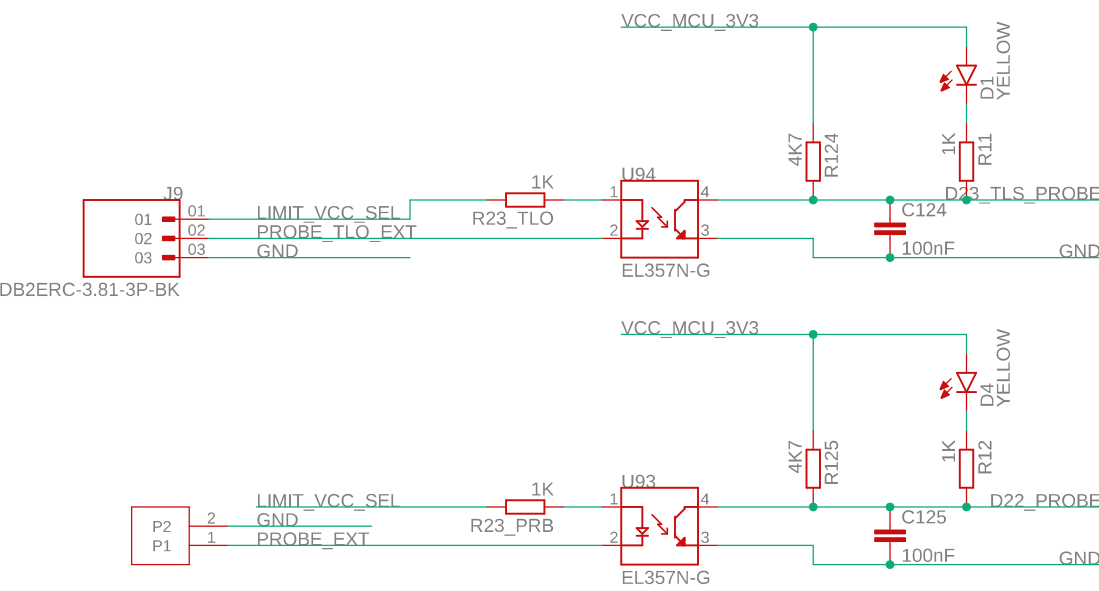
STEP_A_N	7	7	14	A LIM_EXT
STEP_A_P	6	6	13	LIMIT_VCC_SEL
DIR_A_N	5	5	12	GND
DIR_A_P	4	4	11	A_ALARM_5V_N
EN_A_OUT_N	3	3	10	GND
VCC_SYS_5V0	2	2	9	MOTOR_VCC_24-48V
XYA_HOME_EN	1	1	8	GND



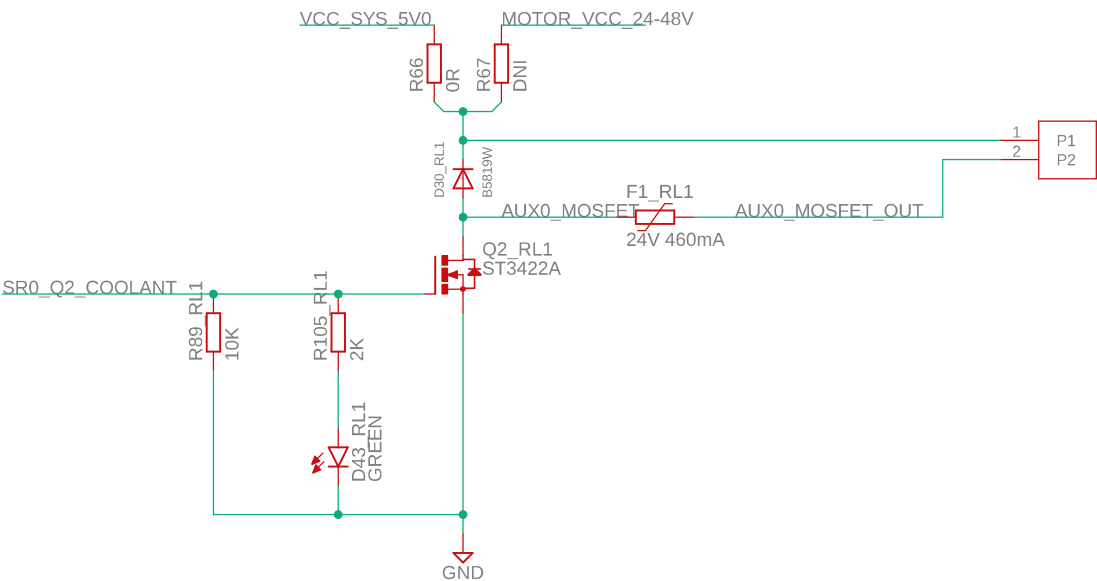
LIMIT SWITCHES



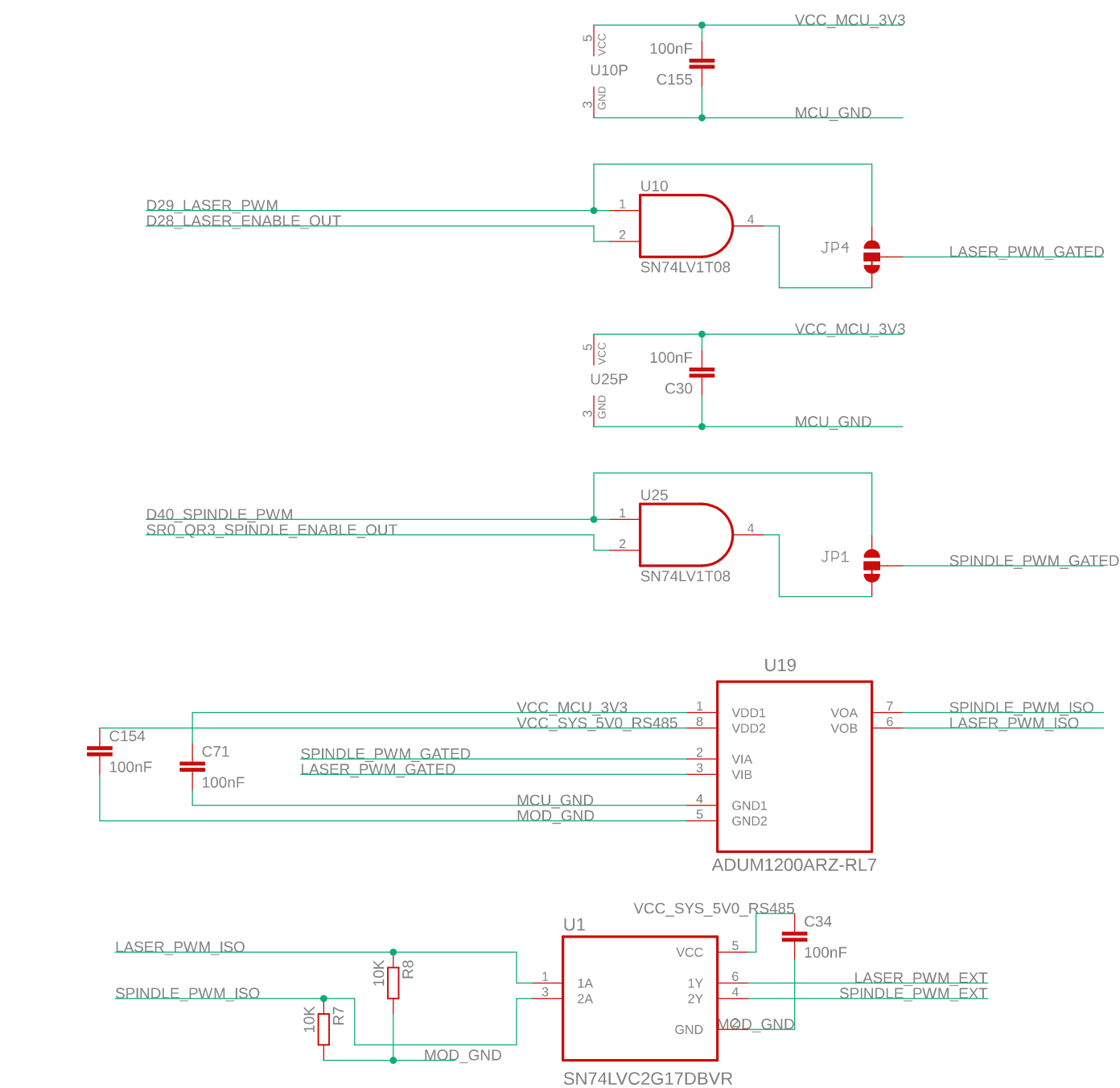
PROBES



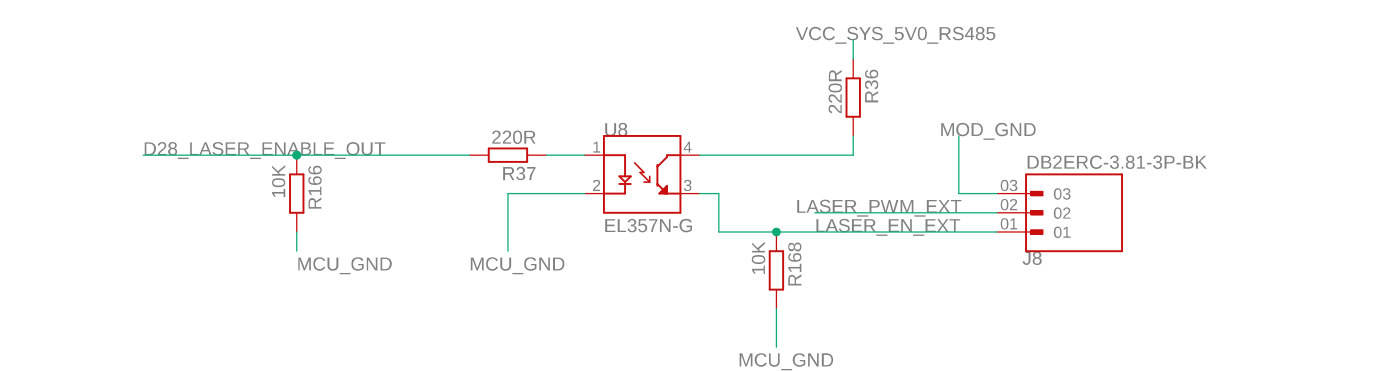
MOSFETS



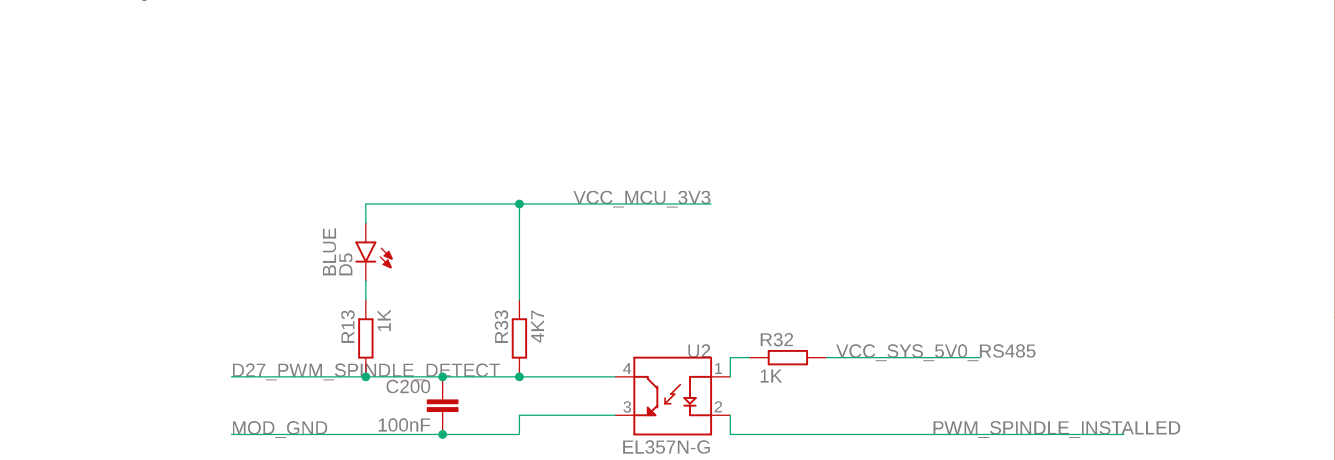
LASER/SPINDLE SAFETY GATING AND ISOLATION



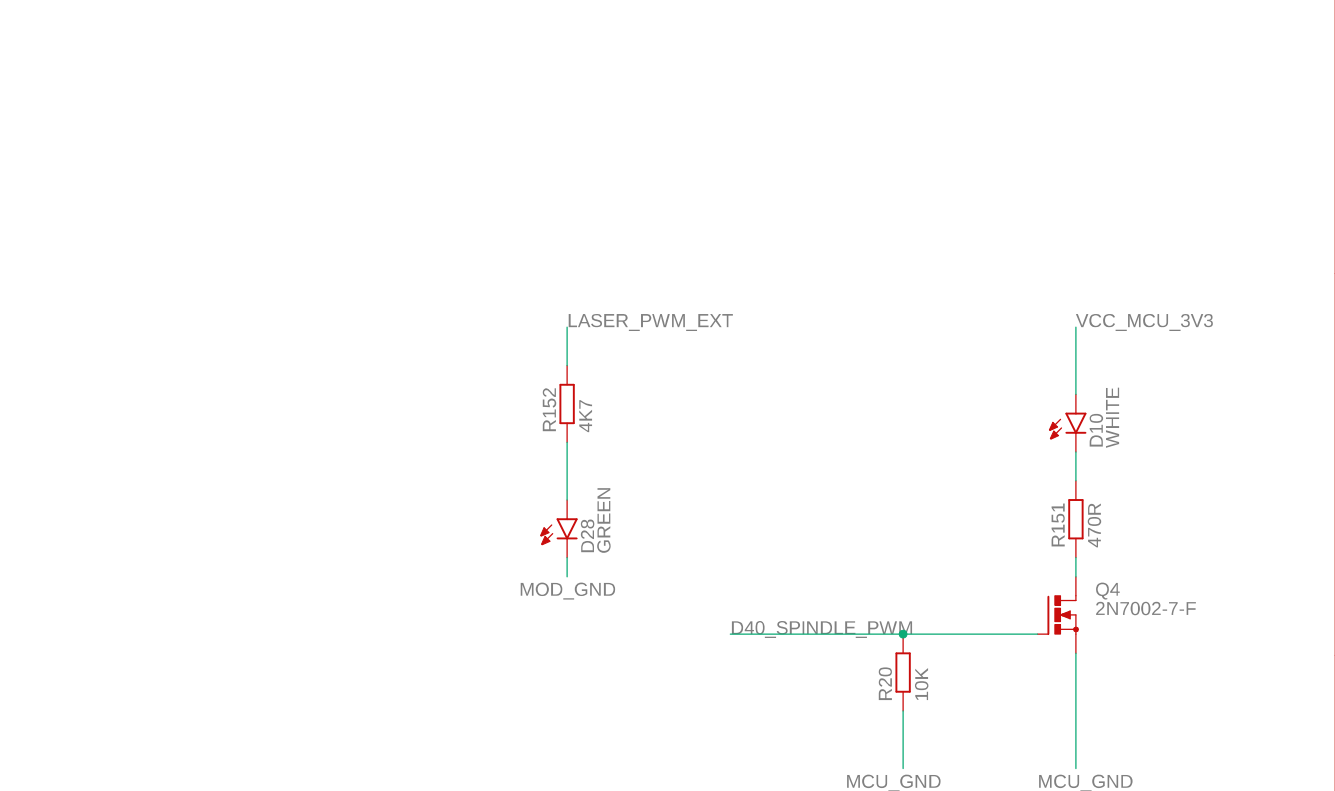
LASER



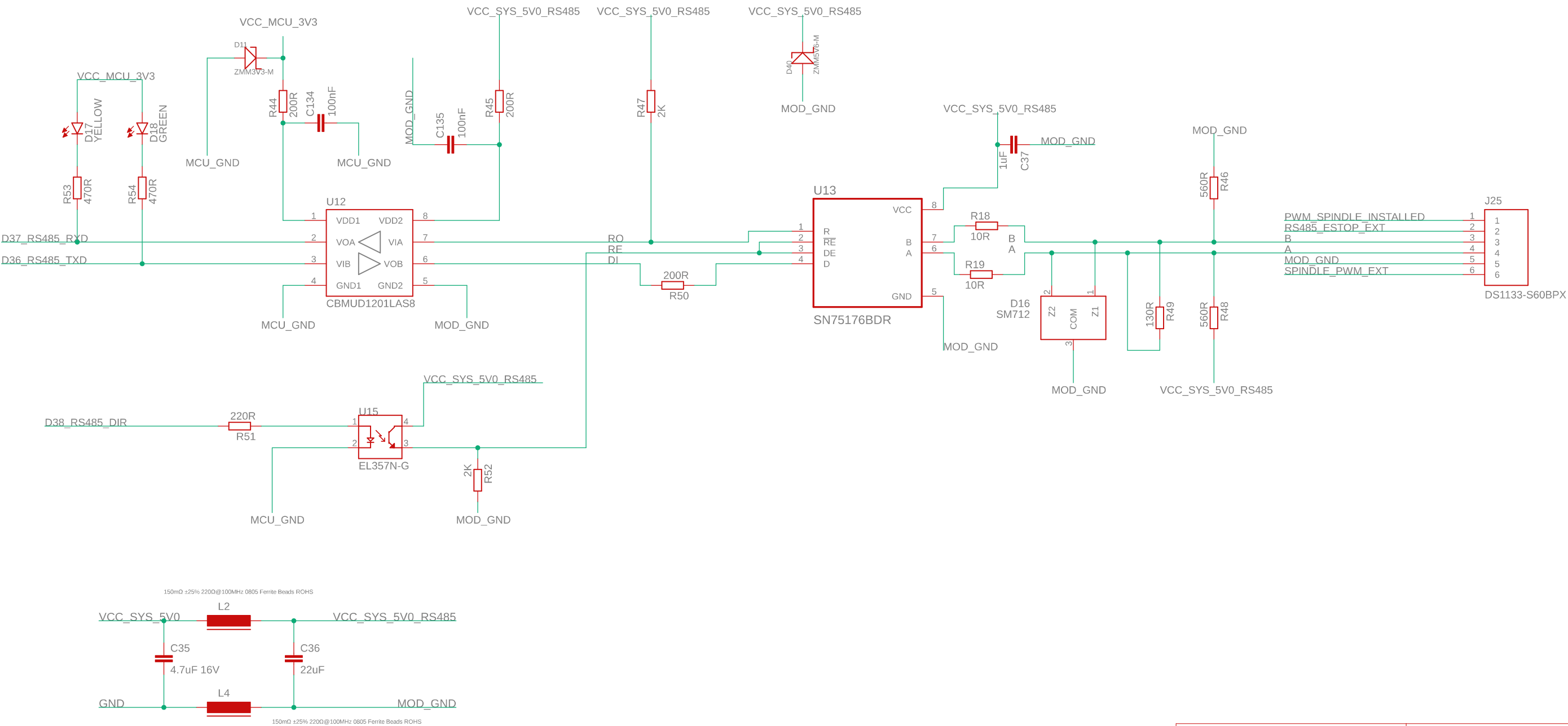
PWM Spindle Detect



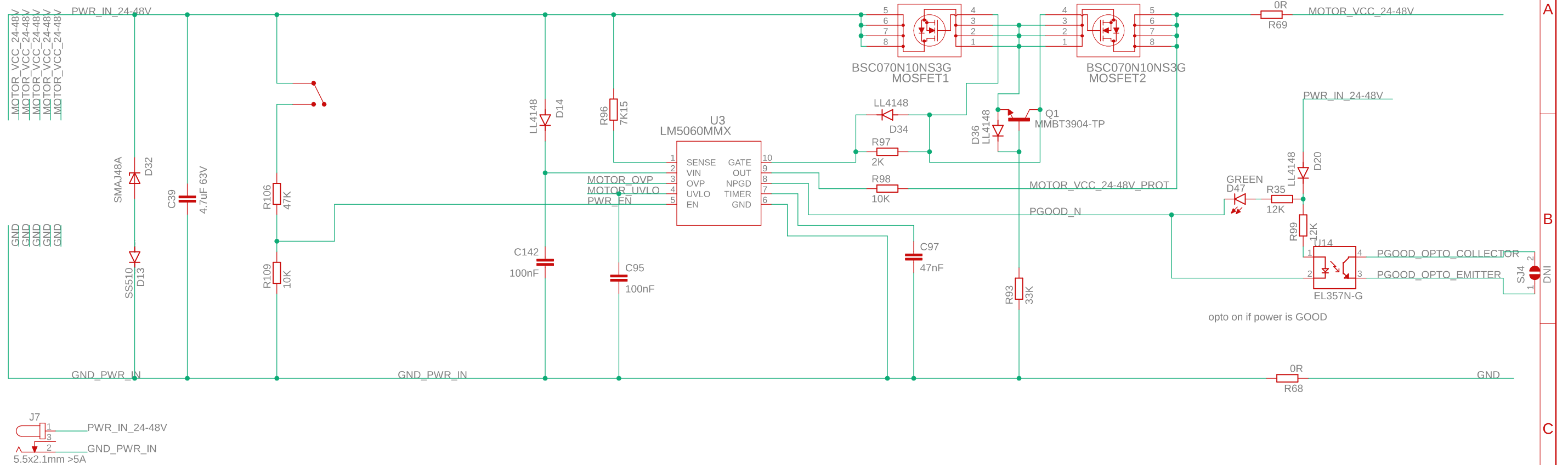
LASER/SPINDLE INDICATOR LEDs



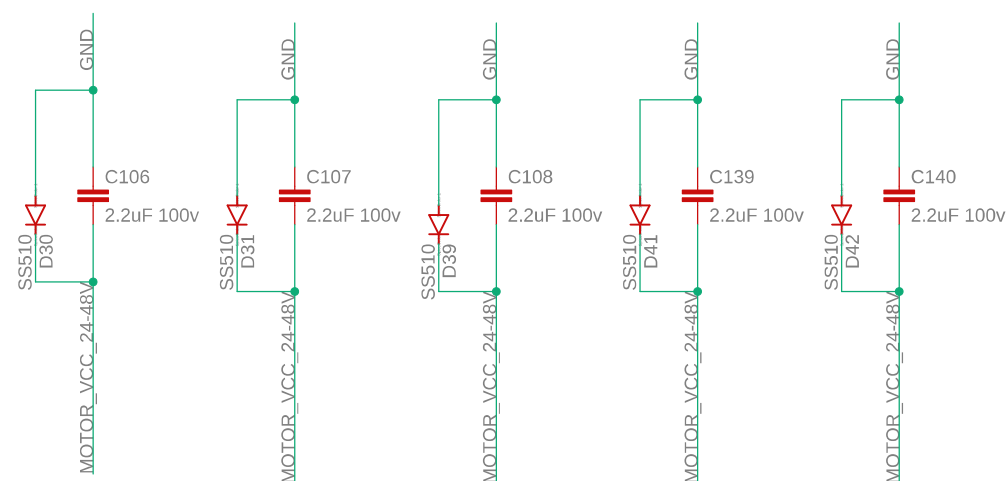
RS-485



eFUSE



PASS THROUGH POWER

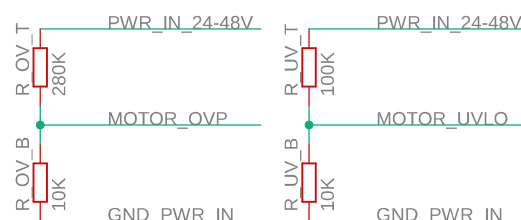


LM5060 configured for 24–48 V input range.

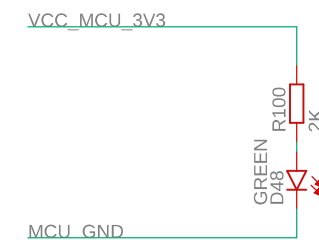
VDS current limit set to ~8 A using $R_S = 7.15 \text{ k}\Omega$.

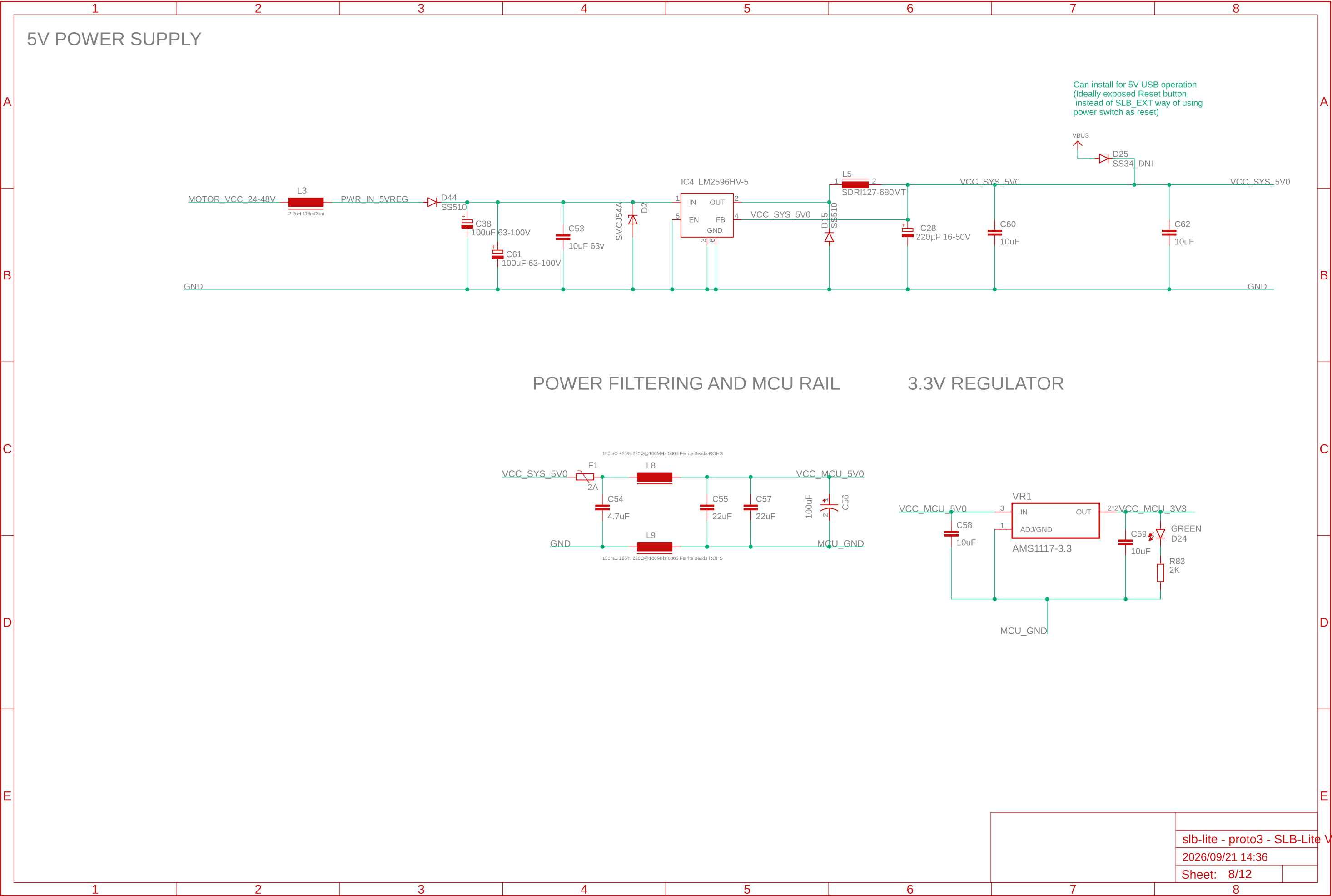
Calculation assumes BSC070N10NS3G MOSFET with ~11 mΩ hot RDS(on):
VDSTH = RS × 16 μA – 25 mV.

UVLO Turn-On 18.15 V
UVLO Turn-Off 15.51 V
OVP Shutdown (Trigger) 58.00 V
OVP Recovery 51.04 V



POWER LED





EMERGENCY STOPS

